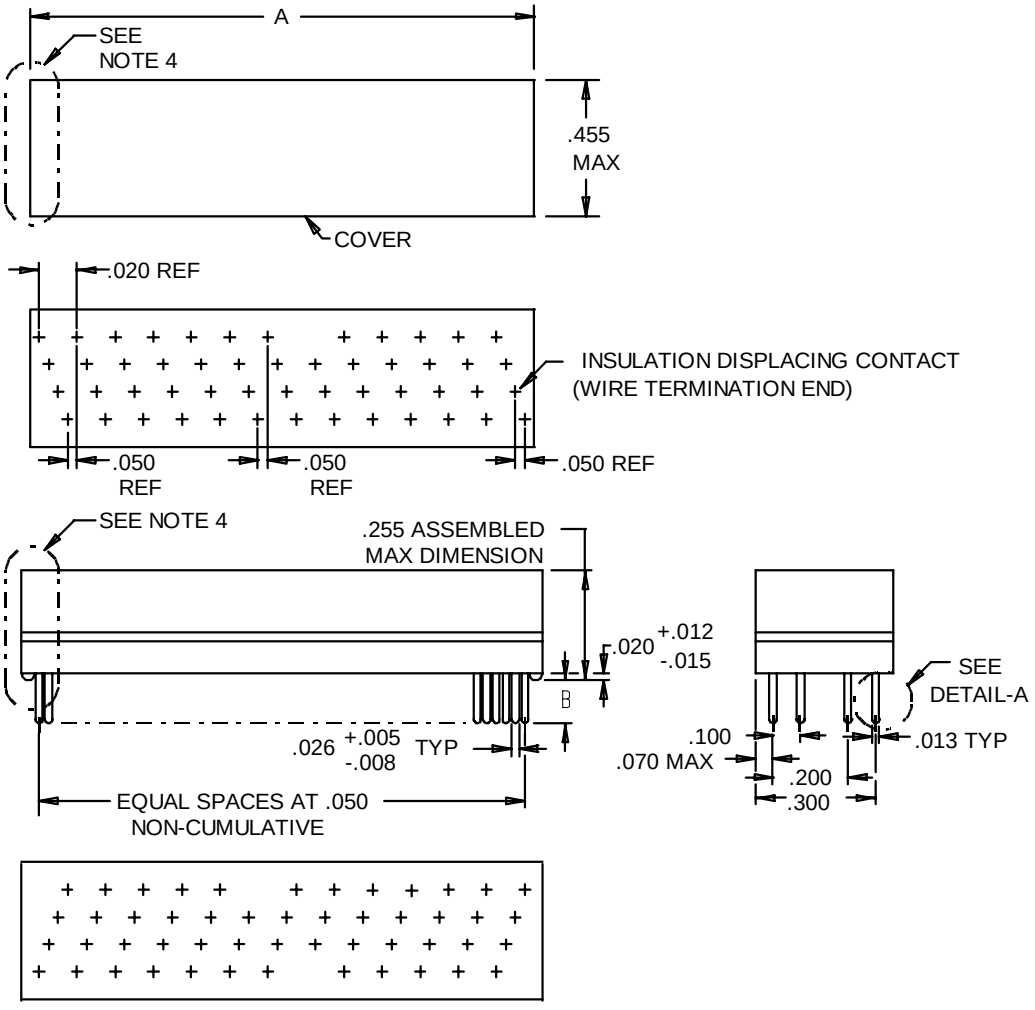
INCH-POUND
MIL-DTL-83503/23B
17 June 1996
SUPERSEDING
MIL-C-83503/23A
1 July 1983

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, FLAT CABLE, NONENVIRONMENTAL PRINTED WIRING BOARD SOLDER TRANSITION FOR TERMINATING ROUND CONDUCTORS ON .050 INCH CENTERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-DTL-83503.



-11 AND -12

(SEE PAGE 2 FOR -01 THRU -10)

Figure 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A Approved for public release; distribution is unlimited.

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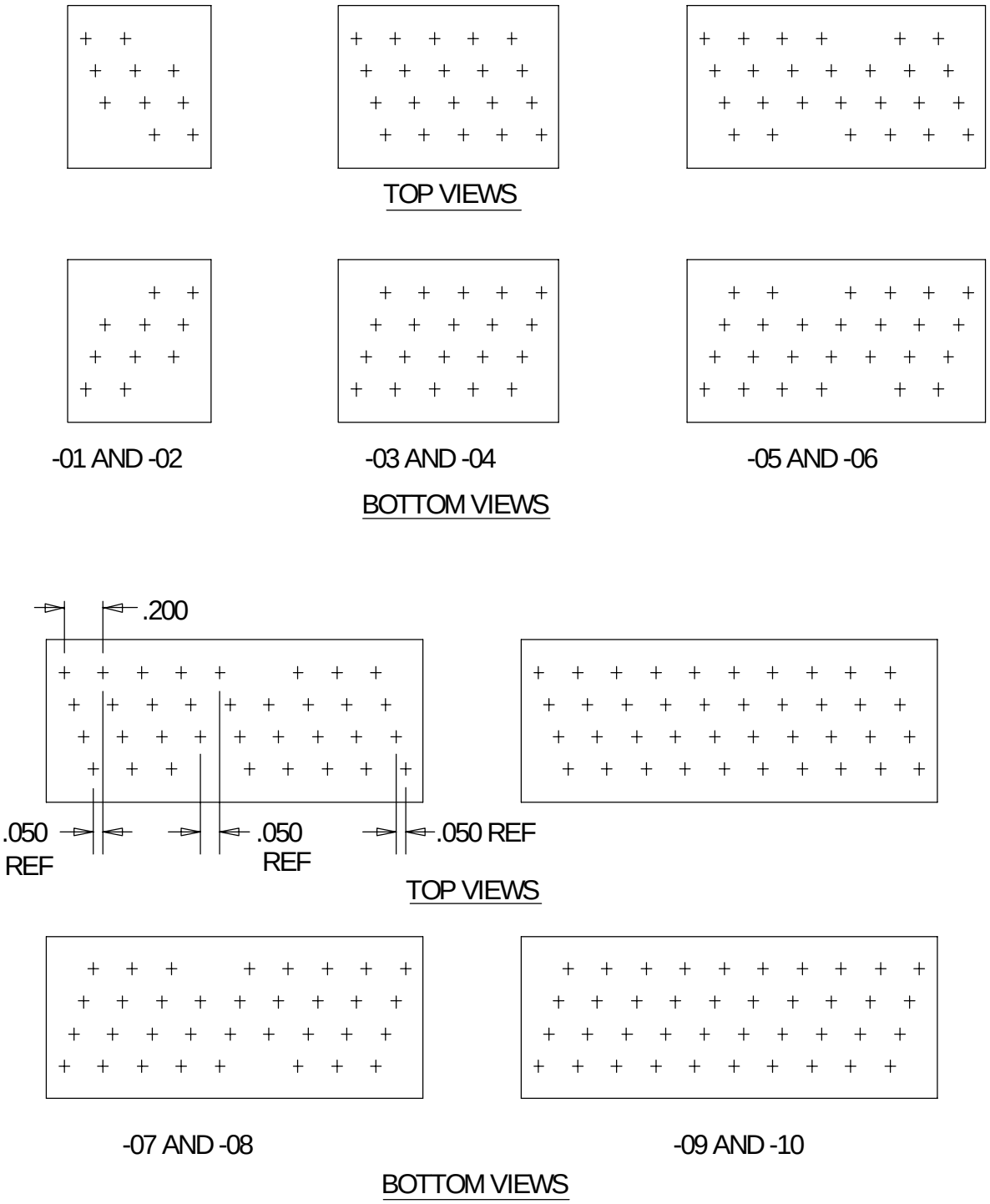

FIGURE 1. Dimensions and configuration - Continued.

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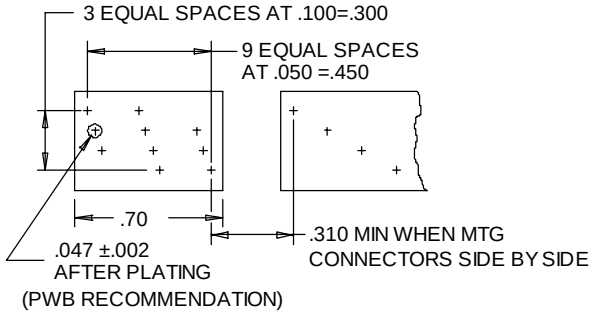
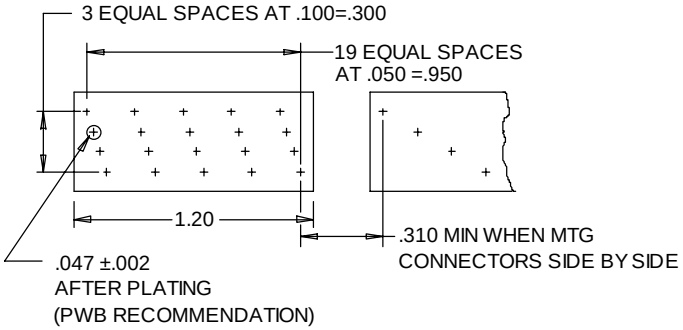
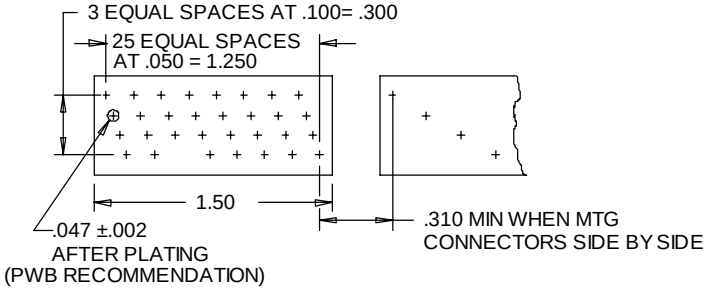


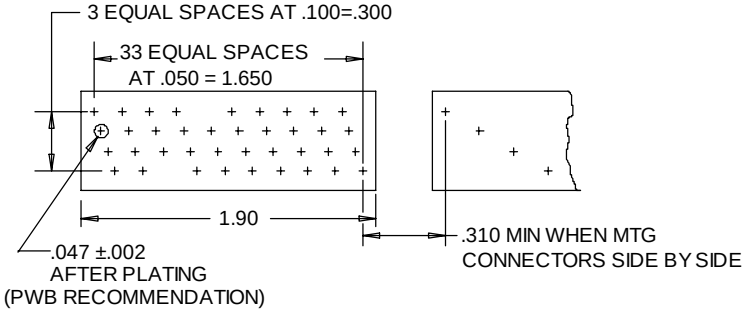
M83503/23 -01 AND -02



M83503/23 -03 AND -04



M83503/23 -05 AND -06



M83503/23 -07 AND -08

FIGURE 1. Dimensions and configuration - Continued.

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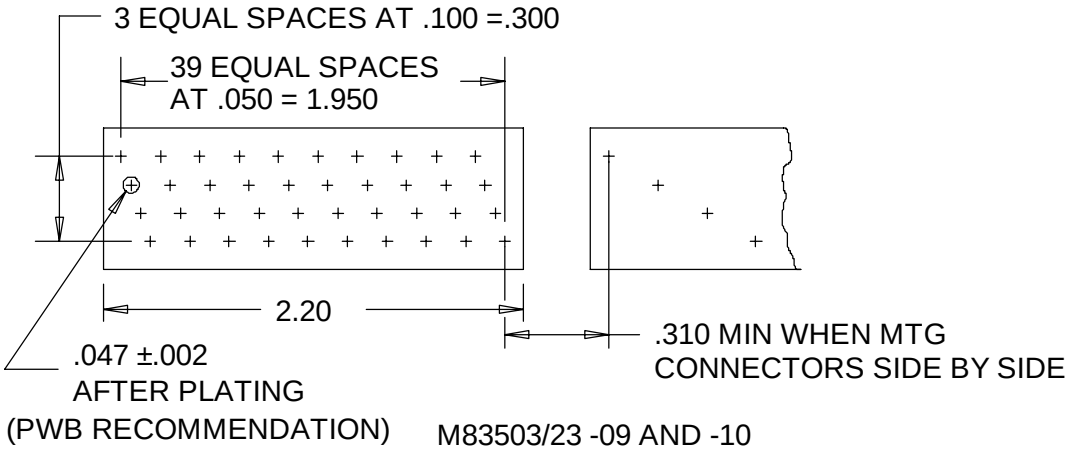
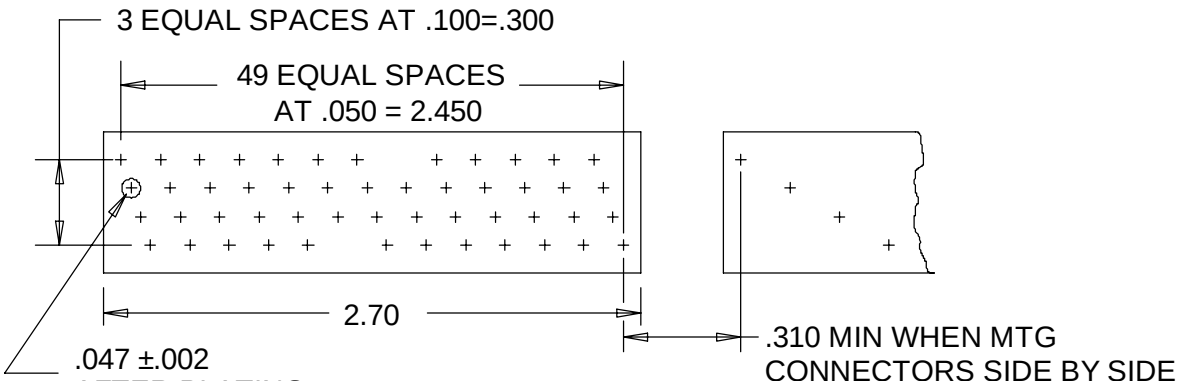


M83503/23 -09 AND -10



M83503/23 -11 AND -12

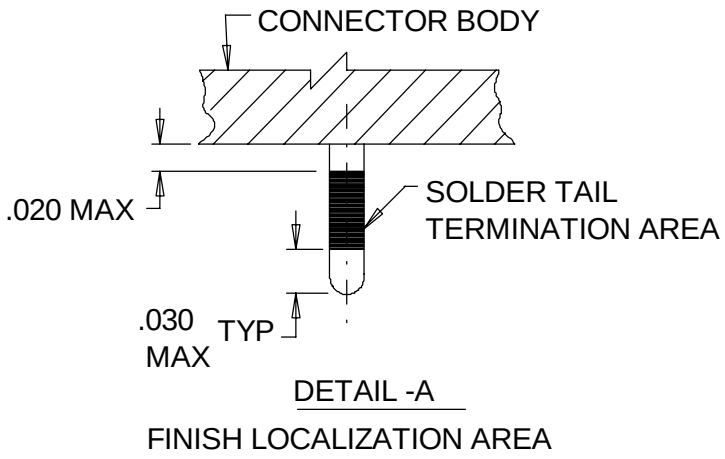


FIGURE 1. Dimensions and configuration - Continued.

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Inches	mm	Inches	mm	Inches	mm
.002	0.05	.100	2.54	1.250	31.75
.010	0.25	.200	5.08	1.50	38.1
.013	0.33	.255	6.48	1.650	41.91
.015	0.38	.300	7.62	1.90	48.3
.020	0.51	.310	7.87	1.950	49.53
.026	0.66	.450	11.43	2.20	55.9
.030	0.76	.455	11.56	2.450	62.23
.047	1.19	.70	17.8	2.70	68.6
.050	1.27	.950	24.13		
.070	1.78	1.20	30.5		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified tolerances are ± .005 (0.13 mm) inch for three place decimals and ± .02 (0.5 mm) inch for two place decimals.
4. Method of connection of cover to body is optional, but shall meet the requirements of MIL-DTL-83503 and this specification sheet.

FIGURE 1. Dimensions and configuration - Continued.

TABLE I. Dash numbers and characteristics

Dash No.	Number of Positions	Dimension	
		A Max	B Max
-01	10	.72 (18.3)	.100 (2.54)
-02	10	.72 (18.3)	.156 (3.96)
-03	20	1.22 (31.0)	.100 (2.54)
-04	20	1.22 (31.0)	.156 (3.96)
-05	26	1.52 (38.6)	.100 (2.54)
-06	26	1.52 (38.6)	.156 (3.96)
-07	34	1.92 (48.3)	.100 (2.54)
-08	34	1.92 (48.3)	.156 (3.96)
-09	40	2.22 (56.4)	.100 (2.54)
-10	40	2.22 (56.4)	.156 (3.96)
-11	50	2.72 (69.1)	.100 (2.54)
-12	50	2.72 (69.1	.156 (3.96)

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REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figure 1 and table I.

Temperature range: -55° to +120°C.

Voltage rating: 300 V rms.

Current rating: 1 ampere.

Materials:

Contacts: Phosphor bronze or brass per CDA260 or ASTM-B-139, spring temper; or beryllium copper alloy CA-725 per ASTM-B-122.

Plating: Tin-lead in the termination area.

Housing: Glass field polyester in accordance with ASTM PS 2, type GPT-15F, GPT-20F, GPT-30F.

Wire accommodation: 28 AWG stranded on .050 (1.27 mm) inch spacing between conductor centerlines.

Voltage rating: 300 volts.

Test wire size: 28 AWG stranded.

Preconditioning: Not applicable.

Dielectric withstanding voltage:
(at sea level) : 500 V rms, minimum.
(at altitude): 200 V rms, minimum.

Cable retention: Not applicable.

Contact resistance: 50 milliohms, maximum.

Mating and unmating force: Not applicable.

Durability: Not applicable.

Interface: Round conductor flat cable, .050 (1.27 mm) inch centers to .062 (1.57 mm) and .093 (2.36 mm) to .125 (3.18 mm) thick printed wiring boards.

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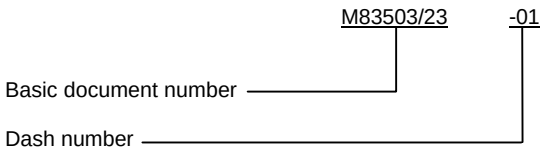
Marking:

Part or identifying number (PIN): The PIN shall consist of this specification sheet and the dash number from table I.

Example



NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes

CONCLUDING MATERIAL

Custodians:
Air Force - 85

Preparing activity:
DLA - CC

(Project 5935-4110-06)

Review activities:
Air Force - 17, 99

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